📘 Introduction to Database & MySQL

Day 1: How Real Apps Store Data

🎯 Level: Very Beginner Friendly | ⏰ Duration: 2 Hours

1. What is Data?

Data = Any information.

If information exists → it is considered data.

Examples of Data:

- Your name & age
- Your exam marks
- Your mobile number
- Product prices on Amazon

2. What is a Database?

A database is a place where data is stored in an **organized way**.

School Register

Stores student details, marks, and attendance records.

Hospital Records

Maintains patient medical history and treatments.

Bank Accounts

Secures financial transactions and balances.

Instagram

Manages user profiles, posts, and followers.

3. DBMS vs. RDBMS

DBMS

Database Management System

Software that provides an interface to:

- Store data
- Retrieve data
- Update & Delete data

Examples: Oracle, SQL Server, MySQL

RDBMS

Relational Database Management System

An advanced structure where:

- Data is strictly stored in **tables**
- Tables can be **connected (related)**

MySQL is an RDBMS.
Tables + Relationships = Real systems.

4. What is MySQL?

MySQL = Open-source database software.

It acts as the structured backbone for modern applications.

Trusted by Tech Giants:

- Facebook
- Twitter
- YouTube

What does it store?

User data, messages, posts, and secure transactions. Whenever you log in somewhere, a database is working!

5. Basic Terms: Columns & Rows

The Structure of a Table

A **Database** is a collection of tables. A **Table** looks like an Excel sheet.

- **Column (Property):** Vertical data representing a specific attribute (e.g., Name, Age).
- **Row (Record):** Horizontal data representing one complete entry (e.g., one student).

id	name	age	city
1	Rahul	18	Delhi
2	Amit	19	Mumbai
3	Sneha	20	Pune

Each row = one student

6. Installation & Connection

Download Required Software

Install both MySQL Server and MySQL Workbench from `mysql.com`.

Set the Root Password

During installation, you will be prompted to set a root password. Memorize this carefully as it is required for access.

Connect to MySQL

Open MySQL Workbench and create a new connection using:

Host: `localhost` | Username: `root` | Password: *[your password]*

Success

If connected successfully, you are officially inside the MySQL environment!

7. Writing First SQL Commands

1. Create & Use Database

First, create a new storage space, then tell the system to use it.

```
CREATE DATABASE first_db;

USE first_db;
```

🎉 **Database created!**

2. Create & Verify Table

Inside the database, define the structure of your data table.

```
CREATE TABLE students (
  id INT,
  name VARCHAR(50),
  age INT
);

SHOW TABLES;
```

8. The MySQL Workflow Concept

1. Database
Create Database
(Like a City)

2. Use DB
Enter Database
(Like a Building)

3. Tables
Create Tables
(Like Rooms)

4. Rows
Store Data
(Like People)

9. Today's Practice Task

> *Do this without copying.*
> *Type it yourself!*
>
> — *The Golden Rule of Coding*

🎯 The Assignment

1. Create a database named `school`.
2. Use the database.
3. Create a table named `teachers`.
4. Add columns:
   `id (INT)`
   `name (VARCHAR)`
   `subject (VARCHAR)`

10. End of Day 1: Concept Checklist

- **What is Data?** (Any information)
- **What is a Database?** (Organized storage)
- **What is DBMS?** (Database Manager)
- **What is RDBMS?** (Relational Data Model)
- **What is MySQL?** (Open-source RDBMS)
- **Tables, Rows, & Columns** (Grid structure)
- **Creating a Database** (`CREATE DATABASE`)
- **Creating a Table** (`CREATE TABLE`)

Tomorrow (Day 2)

- → We will learn Data Types deeply
- → Understanding the Primary Key
- → Inserting real data into tables

Did you actually install MySQL or just read this? 😄

Reply with: 🔥 **"Day 1 completed"**

Image Sources

https://static.vecteezy.com/system/resources/previews/070/719/610/large_2x/abstract-digital-binary-code-stream-data-flow-visualization-in-blue-high-tech-background-photo.jpeg

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